# Milestone 1 User Stories

**Parking Availability Web App**

CS 498 Software Engineering for Senior Project, University of Kentucky, Fall 2026

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## Overview and ownership

This milestone defines 16 user stories for the Parking Availability Web App, with three or four per team member. Each story follows the Agile Alliance template (As a, I want, so that) and carries Given-When-Then acceptance criteria that will become tests. The team has no fixed roles and everyone works across the whole system, so an owner here is only the person accountable for that story. Ownership is spread so each member's stories are in different areas.

**User roles used in these stories**

- **Driver** is anyone checking the lot before arriving. Public, read only, no login.
- **Admin** is a signed-in team member who maps the lot, uploads drone photos, manages closures and watches processing.
- **Developer** is a team member building, testing or deploying the system. Developer stories cover tasks where the user is the builder, not the end user.

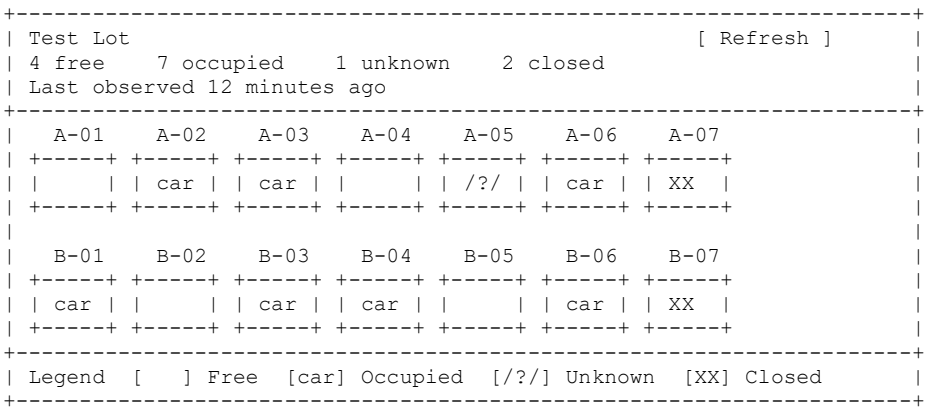
| ID | Story name | Area | Role | Owner |
|---|---|---|---|---|
| US-01 | See Open Spaces at a Glance | Driver dashboard | Driver | Nathan Galante |
| US-02 | Know How Old the Data Is | Driver dashboard | Driver | David Eliassen |
| US-03 | Admin Sign-In | Admin sign-in and uploads | Admin | Rylan Moseley |
| US-04 | Upload a Drone Photo | Admin sign-in and uploads | Admin | Kevin Nguyen |
| US-05 | Block Duplicate Uploads | Admin sign-in and uploads | Admin | Nathan Jones |
| US-06 | Monitor and Retry Processing Jobs | Admin sign-in and uploads | Admin | Nathan Galante |
| US-07 | Draw the Lot Schematic | Lot mapping and closures | Admin | David Eliassen |
| US-08 | Outline Spaces on a Reference Photo | Lot mapping and closures | Admin | Rylan Moseley |
| US-09 | Validate and Publish a Layout | Lot mapping and closures | Admin | Kevin Nguyen |
| US-10 | Close Spaces for an Event | Lot mapping and closures | Admin | Nathan Jones |
| US-11 | Turn a Photo into Space States | Vision pipeline | Driver | Nathan Galante |

| ID | Story name | Area | Role | Owner |
|---|---|---|---|---|
| US-12 | Never Call a Space Free Without Evidence | Vision pipeline | Driver | David Eliassen |
| US-13 | Newer Photos Win | Vision pipeline | Driver | Rylan Moseley |
| US-14 | Run the Whole System on a Laptop | Developer setup | Developer | Kevin Nguyen |
| US-15 | See How Full Each Lot Is | Multiple lots | Driver | Nathan Jones |
| US-16 | Add and Switch Between Lots | Multiple lots | Admin | Nathan Galante |

| Owner | Stories |
|---|---|
| Nathan Galante | US-01, US-06, US-11, US-16 |
| David Eliassen | US-02, US-07, US-12 |
| Rylan Moseley | US-03, US-08, US-13 |
| Kevin Nguyen | US-04, US-09, US-14 |
| Nathan Jones | US-05, US-10, US-15 |

## Driver dashboard

These two stories cover what a driver sees. The public page never shows raw photos and never asks for a login.




*Sketch 1. Driver dashboard*

### US-01 See Open Spaces at a Glance

**Owner** Nathan Galante · **Role** Driver

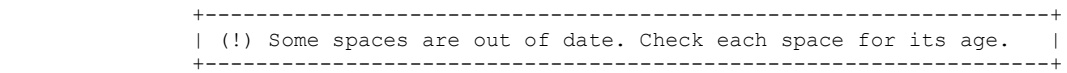
> As a driver, I want to see a simple map of the lot with every space marked open or taken, so that I can decide whether to drive there before I leave.

1. **Dashboard Shows the Schematic**
   - **Given** the lot has a published layout and at least one processed photo,
   - **When** I open the driver page,
   - **Then** I see a white 2D schematic with one rectangle per active space, each labeled with its space ID (Sketch 1), and I am never asked to sign in.
2. **Occupied Spaces Show a Car**
   - **Given** space A-02 is OCCUPIED in the current status,
   - **When** the page loads,
   - **Then** A-02 shows a car icon.
3. **Free Spaces Are Empty**
   - **Given** space A-04 is FREE,
   - **When** the page loads,
   - **Then** A-04 is an empty outline with no icon.
4. **Summary Counts Match the Map**
   - **Given** the lot has spaces in several states,
   - **When** the page loads,
   - **Then** the summary bar shows a count per state that equals the number of rectangles in that state, and closed spaces are never counted as free.
5. **No Photos Yet**
   - **Given** the lot has a published layout but no photo has been processed,
   - **When** I open the driver page,
   - **Then** every nonclosed space shows as UNKNOWN, active closures show as CLOSED, the summary bar says "No observations yet" and the free count is 0.
6. **Page Loads Quickly**
   - **Given** I am on campus Wi-Fi,
   - **When** I open the driver page,
   - **Then** the schematic and counts are visible in under 2 seconds (NFR-2).

## US-02 Know How Old the Data Is

**Owner** David Eliassen · **Role** Driver

> As a driver, I want to see when the lot was last observed, so that I know how much to trust the map before I make the trip.




*Sketch 2. Stale data banner above the summary bar*

1. **Last Observed Time Is Always Shown**
   - **Given** at least one photo has been processed,
   - **When** the page loads,
   - **Then** the summary bar shows "Last observed N minutes ago" based on the newest capture time, not the upload time (NFR-5).

2. **Per-Space Time on Hover or Tap**
    - **Given** I am viewing the dashboard,
    - **When** I hover over or tap space A-03,
    - **Then** a tooltip shows its state and the capture time of the photo behind that state.
3. **Stale Banner Appears**
    - **Given** at least one nonclosed space has an observation older than the lot's allowed data age (for example 30 minutes),
    - **When** the page loads,
    - **Then** a stale-data banner appears, identifies that some spaces are out of date, and greys out only the affected nonclosed spaces. Closed spaces keep their CLOSED indicator.
4. **Fresh Data Clears the Banner**
    - **Given** the stale banner is showing,
    - **When** newer photos finish processing, bring every stale nonclosed space within the allowed data age, and I refresh,
    - **Then** the banner disappears and those spaces return to normal shading. A newer photo that is still older than the allowed age does not clear the warning.
5. **Only Stale Spaces Are Greyed**
    - **Given** spaces seen from viewpoint 1 are fresh and spaces seen only from viewpoint 2 are past the allowed age,
    - **When** the page loads,
    - **Then** only the stale nonclosed viewpoint 2 spaces are greyed, and the banner remains visible until every stale nonclosed space has an observation within the allowed age.

## Admin sign-in and uploads

These four stories cover how an admin signs in, gets photos into the system and keeps processing healthy. Uploads stand in for the fixed cameras a future version would add.

### US-03 Admin Sign-In

**Owner** Rylan Moseley · **Role** Admin

> As an admin, I want to sign in before I can change anything, so that only our team can upload photos, edit layouts or close spaces.

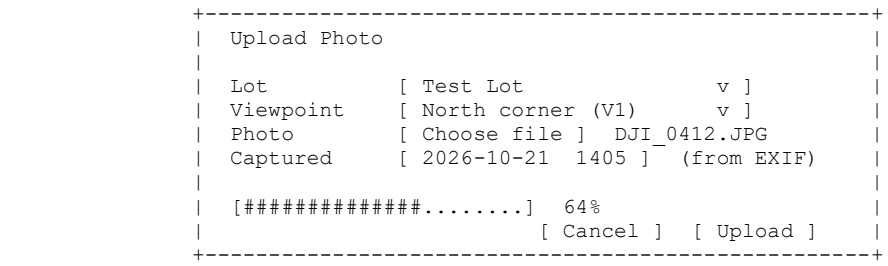
1. **Sign In with Valid Credentials**
    - **Given** I am a member of the admin group,
    - **When** I enter my email and password on the sign-in page,
    - **Then** I land on the admin console home with links to Map, Closures, Upload and Snapshots.
2. **Wrong Password**
    - **Given** I am on the sign-in page,
    - **When** I enter a wrong password,
    - **Then** I see "Email or password is incorrect", I stay on the sign-in page and I am not signed in.

3. **Admin Pages Redirect to Sign-In**
   - **Given** I am not signed in,
   - **When** I open /admin/upload directly,
   - **Then** I am sent to the sign-in page, and after signing in I return to the upload page.
4. **Expired Session**
   - **Given** my session expired while I was on the upload page,
   - **When** I press Upload,
   - **Then** I am sent to the sign-in page with the message "Your session expired. Please sign in again", the photo is not uploaded, and nothing in the lot changes (NFR-6).
5. **Account Without Admin Access**
   - **Given** I sign in with a valid account that is not in the admin group,
   - **When** I open the admin console,
   - **Then** I see "You do not have admin access" and no upload, mapping or closure controls are shown.
6. **Sign Out**
   - **Given** I am signed in,
   - **When** I press Sign out,
   - **Then** I am signed out, I return to the public dashboard, and pressing Back does not show admin pages.

## US-04 Upload a Drone Photo

**Owner** Kevin Nguyen · **Role** Admin

> As an admin, I want to upload a drone photo tagged with its lot, viewpoint and capture time, so that the system can update space availability from it.




*Sketch 3. Upload form*

1. **Capture Time Prefilled from EXIF**
   - **Given** I choose a JPEG that has an EXIF timestamp,
   - **When** the file is selected,
   - **Then** the Captured field fills with that timestamp and I can still edit it.
2. **Missing EXIF Requires a Time**
   - **Given** I choose a photo with no EXIF timestamp,
   - **When** the file is selected,
   - **Then** the Captured field stays empty and Upload is disabled until I enter a time.

3. **Successful Upload Is Queued**
    - **Given** I picked a lot, viewpoint, photo and capture time,
    - **When** I press Upload,
    - **Then** a progress bar fills, I see "Photo queued for processing", and the photo appears in the Snapshots list as QUEUED.
4. **Wrong File Type**
    - **Given** I choose a HEIC or PDF file,
    - **When** the file is selected,
    - **Then** I see "Only JPEG or PNG photos are accepted" and nothing is uploaded.
5. **File Too Large**
    - **Given** I choose a JPEG larger than 15 MB,
    - **When** I press Upload,
    - **Then** I see "Photos must be 15 MB or smaller" and nothing is uploaded.
6. **Capture Time in the Future**
    - **Given** I enter a capture time later than now,
    - **When** I press Upload,
    - **Then** the form shows "Capture time cannot be in the future" and no upload slot is requested.
7. **Abandoned Upload Expires**
    - **Given** I closed the tab before my photo finished uploading,
    - **When** I open the Snapshots list 15 minutes later,
    - **Then** that photo shows as Expired, it is never processed, and the driver map does not change because of it.

## US-05 Block Duplicate Uploads

**Owner** Nathan Jones · **Role** Admin

As an admin, I want the system to reject a photo that was already uploaded, so that one capture is never counted twice and we do not pay to process it again.

1. **Same File Twice**
    - **Given** DJI_0412.JPG was uploaded on Oct 21,
    - **When** I upload the same file again,
    - **Then** I see "This photo was already uploaded on Oct 21" with a link to the original, and no second copy appears in the Snapshots list.
2. **Renamed Copy Is Still a Duplicate**
    - **Given** I renamed that file to lot-north.jpg without changing its contents,
    - **When** I upload it,
    - **Then** it is rejected as a duplicate, because the system compares the photo's contents, not its name.
3. **Edited Photo Is Accepted**
    - **Given** I cropped or re-exported the photo so its bytes changed,
    - **When** I upload it,
    - **Then** it is accepted as a new snapshot.

4. **Two Admins Upload at Once**
   - **Given** two admins upload the same file within the same second,
   - **When** both press Upload,
   - **Then** exactly one upload is accepted and the other admin sees the already uploaded message.

## US-06 Monitor and Retry Processing Jobs

**Owner** Nathan Galante · **Role** Admin

> As an admin, I want to see the state of every uploaded photo and retry the ones that failed, so that a bad run does not leave drivers with old data.

```
+-------------------------------------------------------------------------+
|  Snapshots    Test Lot                                                  |
|                                                                         |
|  Captured      View  State        Time   Model     Error                |
|  Oct 21 1405   V1    COMPLETE     18 s   yolo-s    -                    |
|  Oct 21 1350   V2    PROCESSING   -      yolo-s    -                    |
|  Oct 21 1335   V1    FAILED       -      yolo-s    Too few matches [Retry] |
|  Oct 21 1320   V1    QUEUED       -      -         -                    |
+-------------------------------------------------------------------------+
```

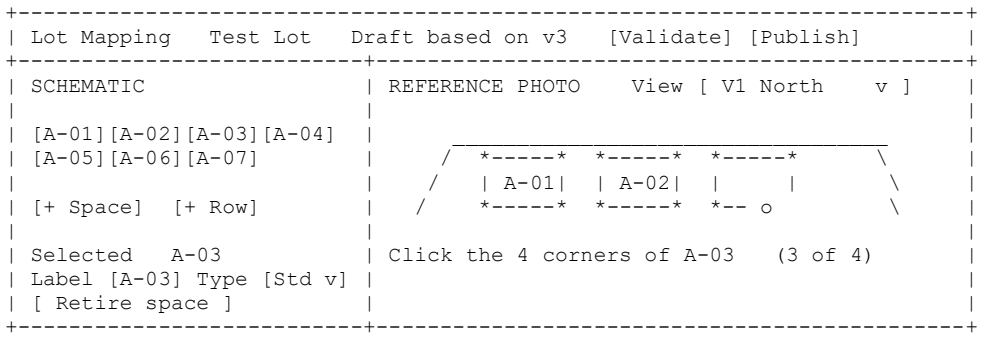
*Sketch 4. Snapshot list*

1. **Recent Snapshots Are Listed**
   - **Given** photos have been uploaded for the lot,
   - **When** I open Snapshots,
   - **Then** I see them newest capture first with viewpoint, job state, processing time, model version and any error (Sketch 4).
2. **Failed Job Shows Its Error**
   - **Given** a job failed because the photo did not match its viewpoint,
   - **When** I view the list,
   - **Then** that row shows FAILED and the error "Too few matches". Failed jobs publish no new observations: prior space states and capture times remain unchanged, spaces with no prior observation stay UNKNOWN, active closures remain CLOSED, and existing observations still age under US-02.
3. **Retry a Failed Job**
   - **Given** a snapshot is FAILED,
   - **When** I press Retry,
   - **Then** its state changes to QUEUED, then to COMPLETE or FAILED, and the driver view updates only if processing completes successfully and that photo is the newest for its spaces.
4. **Retry Only Where It Makes Sense**
   - **Given** a snapshot is COMPLETE, QUEUED or PROCESSING,
   - **When** I view its row,
   - **Then** no Retry button is shown.
5. **Repeated Failures Stop Automatically**
   - **Given** a photo has failed three times in a row,
   - **When** I view the Snapshots list,
   - **Then** it shows FAILED with "Failed after 3 attempts", the system stops retrying it on its own, and my Retry button is still available.

6. **Snapshot Detail**
   - **Given** a snapshot is COMPLETE,
   - **When** I open it,
   - **Then** I see each space's state, confidence and overlap from that photo, plus the layout version it was processed with.

## Lot mapping and closures

These four stories cover how an admin turns a physical lot into data and controls what drivers see. The draft layout is the only place a lot is edited, and only a published version reaches photo processing and the driver map.




*Sketch 5. Lot mapping page, with the schematic on the left and the reference photo on the right*

### US-07 Draw the Lot Schematic

**Owner** David Eliassen · **Role** Admin

> As an admin, I want to draw the lot as a simple map of labeled spaces, so that drivers see a clean schematic instead of a raw photo.

1. **Add a Single Space**
   - **Given** I am on the mapping page for Test Lot,
   - **When** I press + Space,
   - **Then** a rectangle appears with the next unused label and is saved to the draft.
2. **Add a Whole Row**
   - **Given** I am on the mapping page,
   - **When** I press + Row, enter 10 spaces starting at B-01 and confirm,
   - **Then** ten evenly spaced rectangles labeled B-01 through B-10 appear in a line.
3. **Move and Resize**
   - **Given** space A-03 is on the schematic,
   - **When** I drag it to a new spot and resize it,
   - **Then** the new position is saved to the draft and the driver view does not change until I publish.
4. **Set Label and Type**
   - **Given** I selected A-03,
   - **When** I change its type to ADA or EV,
   - **Then** the type is saved with the space and shown on it in the editor.

5. **Duplicate Label Warning**
    - **Given** B-04 already exists,
    - **When** I rename another space to B-04,
    - **Then** both spaces are outlined in red with the note "Label already used".
6. **Retire a Space**
    - **Given** a space was painted over and no longer exists,
    - **When** I press Retire space,
    - **Then** it is greyed out in the editor, disappears from the driver view after the next publish, and its past observations stay in history.
7. **Draft Survives a Reload**
    - **Given** I made edits but have not published,
    - **When** I reload the page or sign in from another laptop,
    - **Then** my edits are still in the draft.

## US-08 Outline Spaces on a Reference Photo

**Owner** Rylan Moseley · **Role** Admin

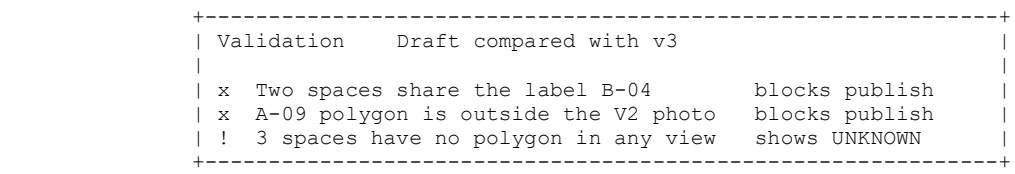
As an admin, I want to click the corners of each space on a reference photo for each drone viewpoint, so that the app knows which part of a new photo belongs to which space.

1. **Upload a Reference Photo**
    - **Given** viewpoint V1 has no reference photo,
    - **When** I upload one for V1,
    - **Then** it appears in the right panel of the mapping page (Sketch 5).
2. **Outline a Space**
    - **Given** I selected A-03 on the schematic and the V1 photo is showing,
    - **When** I click four corners on the photo,
    - **Then** a polygon is drawn, linked to A-03, and the panel reads "A-03 outlined in V1".
3. **Adjust a Polygon**
    - **Given** A-03 already has a polygon in V1,
    - **When** I drag one of its corner handles,
    - **Then** the polygon updates and the change is saved to the draft.
4. **One Space in Two Viewpoints**
    - **Given** A-03 is visible from both V1 and V2,
    - **When** I outline it in each view,
    - **Then** both polygons are stored and A-03 shows a count of 2 views.
5. **Cancel an Outline Midway**
    - **Given** I clicked two of four corners,
    - **When** I press Esc,
    - **Then** the partial outline is discarded and nothing is saved.
6. **Linked Highlight**
    - **Given** spaces are outlined in V1,
    - **When** I hover over A-03 on the schematic,
    - **Then** its polygon is highlighted on the photo, and hovering the polygon highlights the rectangle.

## US-09 Validate and Publish a Layout

**Owner** Kevin Nguyen · **Role** Admin

> As an admin, I want to check my draft for mistakes and publish it as a numbered version, so that photo processing and the driver map always use the same layout and I can go back if something is wrong.



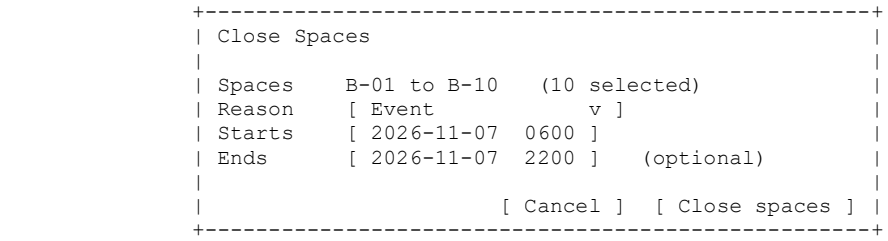

*Sketch 6. Validation panel*

1. **Validate Lists Problems**
   - **Given** my draft has duplicate labels, an out-of-bounds polygon and unmapped spaces,
   - **When** I press Validate,
   - **Then** I see each problem marked as an error (x) or a warning (!) as in Sketch 6.
2. **Errors Block Publishing**
   - **Given** validation found at least one error,
   - **When** I look at the Publish button,
   - **Then** it is disabled until every error is fixed.
3. **Warnings Do Not Block**
   - **Given** validation found only warnings,
   - **When** I press Publish,
   - **Then** the layout publishes and the unmapped spaces show as UNKNOWN to drivers.
4. **Publish Creates a New Version**
   - **Given** the lot is on v3 and my draft is valid,
   - **When** I press Publish,
   - **Then** v4 is saved with my name and the time, the lot points to v4, drivers see it on their next refresh, and the next photo is processed with v4.
5. **Photos in Progress Keep Their Version**
   - **Given** a photo was still processing when I published v4,
   - **When** I open that photo in the Snapshots list,
   - **Then** it shows it was processed with v3, and its results only cover spaces that existed in v3.
6. **Version History**
   - **Given** the lot has several published versions,
   - **When** I open Versions,
   - **Then** I see each version number with who published it and when, newest first.
7. **Roll Back to an Earlier Version**
   - **Given** v4 has a mistake,
   - **When** I choose Restore on v3 and publish,
   - **Then** a new v5 is created with the same contents as v3, and v4 stays in history unchanged.

## US-10 Close Spaces for an Event

**Owner** Nathan Jones · **Role** Admin

> As an admin, I want to block off spaces for construction, events or maintenance for a set time, so that drivers do not head to a space they cannot use.




*Sketch 7. Close spaces dialog*

1. **Close a Row**
   - **Given** I selected row B on the schematic,
   - **When** I choose Event, set a start and end time that includes now, and press Close spaces,
   - **Then** on the next driver refresh B-01 through B-10 show as CLOSED with a barrier icon and are left out of the free count.
2. **Closure Beats Detection**
   - **Given** B-03 is closed and the newest photo shows it empty,
   - **When** a driver loads the page,
   - **Then** B-03 shows CLOSED, not FREE.
3. **Scheduled Closure Starts on Its Own**
   - **Given** I created a closure that starts tomorrow at 0600,
   - **When** a driver loads the page before then,
   - **Then** those spaces show their normal state, and after 0600 they show CLOSED with no one taking action.
4. **Closure Ends on Its Own**
   - **Given** a closure's end time has passed,
   - **When** a driver refreshes,
   - **Then** the spaces show their detected state again.
5. **End a Closure Early**
   - **Given** a closure is active,
   - **When** I press End now on the Closures page,
   - **Then** the spaces return to their detected state on the next refresh.
6. **Open-Ended Closure**
   - **Given** I leave the end time empty,
   - **When** I save the closure,
   - **Then** the spaces stay CLOSED until an admin ends it.
7. **End Before Start Is Rejected**
   - **Given** I enter an end time earlier than the start time,
   - **When** I press Close spaces,
   - **Then** I see "End time must be after start time" and nothing is saved.

8. **Closures Are Recorded**
    - **Given** closures exist for the lot,
    - **When** I open the Closures page,
    - **Then** each one lists its spaces, reason, time window, who created it and when.

## Vision pipeline

These three stories cover what happens between an upload and a change on the map. All three are written from the driver's side because the driver is who feels a wrong result.

### US-11 Turn a Photo into Space States

**Owner** Nathan Galante · **Role** Driver

> As a driver, I want each new drone photo turned into free, occupied or unknown for every space it covers, so that the map matches what the lot actually looks like.

1. **Clear Photo Updates the Map**
    - **Given** an admin uploaded a clear photo from viewpoint V1 with cars in A-02, A-03 and A-06,
    - **When** I refresh the driver page after it finishes processing,
    - **Then** those three spaces show car icons and the other V1 spaces show as free.
2. **Car Parked Over a Line**
    - **Given** a car is parked mostly in A-04 and slightly over the line into A-05,
    - **When** I view the map after that photo is processed,
    - **Then** A-04 shows as occupied and A-05 shows as unknown rather than free.
3. **Spaces Outside the Photo Keep Their State**
    - **Given** the newest photo only shows row A,
    - **When** I view the map,
    - **Then** every row B space keeps its earlier state and its earlier observed time.
4. **Results Arrive Within a Minute**
    - **Given** an admin just finished uploading a photo,
    - **When** I refresh the driver page within a minute,
    - **Then** the map shows that photo's results (NFR-1).
5. **Processing Never Blanks the Map**
    - **Given** a new photo is still processing,
    - **When** I refresh the driver page,
    - **Then** I still see the previous results with their observed time, plus a small "Processing new photo" note.

### US-12 Never Call a Space Free Without Evidence

**Owner** David Eliassen · **Role** Driver

> As a driver, I want the app to say "unknown" when it is not sure, so that I never drive to a space the app wrongly called free.

1. **Photo Does Not Match Its Viewpoint**
   - **Given** an admin uploaded a photo tagged V1 that was actually taken from a different spot,
   - **When** processing rejects that photo and I view the map,
   - **Then** the rejected photo contributes no new observations. Each V1 space keeps its prior state and observed time, or shows UNKNOWN if it has no prior observation; active closures remain CLOSED. No space is marked free because of the rejected photo, and stale prior observations retain the US-02 warning.
2. **Space Outside the Frame**
   - **Given** the drone drifted so A-01 is cut off at the photo edge,
   - **When** I view the map after the photo is processed,
   - **Then** A-01 shows as unknown while the fully visible spaces show normal results.
3. **Hard-to-See Space**
   - **Given** A-06 is in heavy shade so the app cannot tell whether a car is there,
   - **When** I view the map,
   - **Then** A-06 shows as unknown, not free and not occupied.
4. **Unknown Never Counts as Free**
   - **Given** three spaces are unknown,
   - **When** I load the page,
   - **Then** none of the three are included in the free count.
5. **A Wrong Free Is Rare**
   - **Given** the team's set of labeled test photos,
   - **When** those photos are run through the app,
   - **Then** no more than 5 percent of spaces that actually have a car show as free on the map (NFR-3).

## US-13 Newer Photos Win

**Owner** Rylan Moseley · **Role** Driver

As a driver, I want every space to show its newest observation, so that a slow or late photo never replaces fresher information.

1. **Late Older Photo Does Not Overwrite**
   - **Given** A-03 shows a result from a photo captured at 1405,
   - **When** an older photo captured at 1350 finishes processing afterward,
   - **Then** A-03 still shows the 1405 result and the 1405 observed time.
2. **Capture Time Decides, Not Upload Time**
   - **Given** a photo captured at 1320 was uploaded after a photo captured at 1335,
   - **When** both finish processing,
   - **Then** the map shows the 1335 results.
3. **Same Photo Processed Twice**
   - **Given** the system accidentally processes the same photo twice,
   - **When** I refresh the map,
   - **Then** nothing changes and the observed time stays the same.

4. **Two Viewpoints Finish in Either Order**
    - **Given** A-03 is in both V1 (captured 1410) and V2 (captured 1405),
    - **When** the V1 photo finishes processing before the V2 photo,
    - **Then** A-03 shows the V1 result.

## Developer setup

This story is developer-centric. The user is a teammate building and testing the system, and the goal is the same code running on a laptop in Phase 1 and on AWS in Phase 2.

### US-14 Run the Whole System on a Laptop

**Owner** Kevin Nguyen · **Role** Developer

> As a developer on the team, I want to start the whole system with one command, so that I can build and test features on my own laptop without an AWS account.

```
$ cp .env.example .env
$ docker compose up --build
  web             running    localhost port 5173
  api             running    localhost port 8000    /health ok
  worker          running    polling queue
  minio           running    localhost port 9000
  elasticmq       running    localhost port 9324
  dynamodb-local  running    localhost port 8001
$ python seed/seed.py --lot test-lot
  Created table, bucket and queue
  Loaded Test Lot with 14 spaces and 2 viewpoints
```

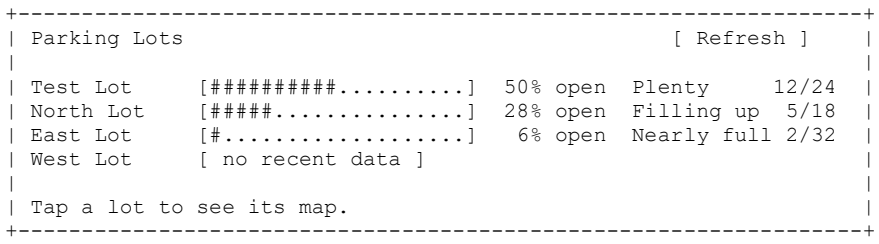
*Sketch 8. Local startup from a clean clone*

1. **Clean Clone Starts**
    - **Given** a fresh clone on a laptop with Docker installed,
    - **When** I copy .env.example to .env and run docker compose up --build,
    - **Then** all six services start and GET /health returns ok.
2. **Seed Loads the Test Lot**
    - **Given** the stack is running,
    - **When** I run the seed script,
    - **Then** the table, bucket and queue are created and Test Lot appears on the dashboard at localhost port 5173.
3. **Upload to Map, Fully Local**
    - **Given** the stack is seeded,
    - **When** I sign in as the seeded dev admin and upload a sample photo from eval/,
    - **Then** the schematic updates with that photo's results.
4. **No Cloud Credentials Needed**
    - **Given** my laptop has no AWS credentials,
    - **When** I run the steps above,
    - **Then** everything works because MinIO, ElasticMQ and DynamoDB Local stand in for AWS.

5. **Only Configuration Differs**
    - **Given** APP_ENV is set to local,
    - **When** a reviewer reads the API and worker code,
    - **Then** local or AWS behavior is chosen only by environment variables, with no environment-specific branches in application code (NFR-7).
6. **Seeding Twice Is Safe**
    - **Given** I already seeded the stack,
    - **When** I run the seed script again,
    - **Then** it finishes without errors and creates no duplicate lots or spaces.

## Multiple lots

These two stories extend the app from one lot to several. Drivers pick a lot from an overview that shows how full each one is, and admins add lots and switch between them with the same mapping, upload and closure tools.




*Sketch 9. Lot overview list*

### US-15 See How Full Each Lot Is

**Owner** Nathan Jones · **Role** Driver

> As a driver, I want a list of lots that shows how full each one is, so that I can pick a lot at a glance and then open its detailed map.

1. **Lot List Appears**
    - **Given** there are published lots,
    - **When** I open the driver home page,
    - **Then** I see every published lot with its name, a progress bar, the percent of open spaces and the open count, for example "12/24" (Sketch 9), and I am never asked to sign in.
2. **Color and Label Show How Full**
    - **Given** a lot's percent of open spaces is known,
    - **When** I view the list,
    - **Then** its bar is green with the label Plenty at 30 percent or more open, yellow with Filling up from 10 to 29 percent, and red with Nearly full under 10 percent, so the meaning never depends on color alone.

3. **Only Free Spaces Count as Open**
   - **Given** North Lot has 20 spaces with 5 free, 10 occupied, 3 unknown and 2 closed,
   - **When** I view the list,
   - **Then** North Lot shows 5/18 open and 28 percent, because closed spaces leave the total and unknown spaces never count as open.
4. **Some Data Out of Date**
   - **Given** some nonclosed spaces in North Lot are past the allowed data age,
   - **When** I view the list,
   - **Then** North Lot still shows its bar with a "Some data out of date" note next to it.
5. **No Recent Data**
   - **Given** no nonclosed space in West Lot has an observation within the allowed data age,
   - **When** I view the list,
   - **Then** West Lot shows a grey "no recent data" bar with no percentage.
6. **Open a Lot**
   - **Given** I am on the lot list,
   - **When** I tap North Lot,
   - **Then** I see North Lot's map from US-01, and pressing Back returns me to the list.
7. **List Matches the Map**
   - **Given** I opened North Lot from the list,
   - **When** I compare the two screens,
   - **Then** the open count on the list equals the free count in North Lot's summary bar.

## US-16 Add and Switch Between Lots

**Owner** Nathan Galante · **Role** Admin

As an admin, I want to add another parking lot and switch between lots, so that the app can cover more than one lot with the same tools.

1. **Create a Lot**
   - **Given** I am signed in as an admin,
   - **When** I press New Lot and enter a name, a location and an allowed data age,
   - **Then** the lot appears in the admin lot picker with an empty draft layout.
2. **Duplicate Lot Name**
   - **Given** Test Lot already exists,
   - **When** I create another lot named Test Lot,
   - **Then** I see "A lot with this name already exists" and nothing is created.
3. **Lot Picker on Every Admin Page**
   - **Given** Test Lot and North Lot both exist,
   - **When** I choose North Lot in the lot picker,
   - **Then** the mapping, upload, closures and snapshots pages show only North Lot's data until I switch again.

4. **Unpublished Lot Stays Hidden**
    - **Given** North Lot has no published layout yet,
    - **When** a driver opens the lot list,
    - **Then** North Lot does not appear.
5. **Published Lot Appears for Drivers**
    - **Given** I publish North Lot's first layout,
    - **When** a driver refreshes the lot list,
    - **Then** North Lot appears with its own bar, and its spaces show as unknown until a photo covers them.
6. **Lots Stay Separate**
    - **Given** row B of Test Lot is closed for an event,
    - **When** a driver opens North Lot,
    - **Then** none of North Lot's spaces are closed and its counts are unaffected.

## Referenced nonfunctional requirements

This key defines the NFR labels referenced in these stories. It retains the existing numbering; only requirements cited in this submission are listed.

NFR-1 — Processing latency (US-11). At least 95 percent of valid benchmark photos have results available to the driver page within 60 seconds of upload completion. Include queue wait and processing time. Record the sample size, photo dimensions and sizes, lot and viewpoint coverage, server hardware, model version, concurrent upload load and measurement method so the test can be repeated.

NFR-2 — Dashboard load time (US-01). On campus Wi-Fi, the schematic and summary counts are visible in under 2 seconds from navigation to the driver page. Record the device, browser, network conditions, lot size, cache state and number of test runs; report the measured load times.

NFR-3 — False-free rate (US-12). On the team's labeled evaluation photos, no more than 5 percent of actually occupied space observations may be classified FREE. Compute the rate as occupied observations classified FREE divided by all labeled occupied observations, multiplied by 100. Report both counts and the evaluation set and model versions; the set must contain occupied spaces.

NFR-5 — Observation freshness (US-02). Observation age uses capture time, not upload or processing time. Evaluate staleness per space against the lot's configured allowed age. A banner remains while any nonclosed space with an observation exceeds that age. Spaces without observations show UNKNOWN; active closures retain CLOSED. Failed jobs do not advance observation times.

NFR-6 — Expired-session protection (US-03). An expired session cannot upload a photo or change lot data. The admin must sign in again and receives the session-expired message specified in US-03.

NFR-7 — Local/cloud configuration (US-14). The API and worker select local or AWS services through environment configuration, without environment-specific branches in application code. The documented local workflow runs without AWS credentials.